

Lab 37: Data Cleaning with Pandas (Missing Values & Duplicates)

Aim

To identify and handle **missing values and duplicate records** in a dataset using Pandas.

Algorithm

1. Import the Pandas library.
2. Create a DataFrame containing missing and duplicate values.
3. Identify missing values using `isnull()`.
4. Fill missing numerical values using the mean.
5. Identify duplicate records.
6. Remove duplicate rows using `drop_duplicates()`.
7. Display the cleaned DataFrame.

Program

Lab 37: Data Cleaning with Pandas (Missing Values & Duplicates)

```
import pandas as pd
import numpy as np

# Create DataFrame with missing values and duplicates
df = pd.DataFrame({
    "name": ["A", "B", "C", "B", "D"],
    "marks": [78, np.nan, 90, np.nan, 85]
})

print("Original DataFrame:\n", df)

# Check missing values
print("\nMissing Values:\n", df.isnull().sum())
```

```
# Fill missing values with mean
```

```
df["marks"] = df["marks"].fillna(df["marks"].mean())
```

```
# Remove duplicate rows
```

```
df = df.drop_duplicates()
```

```
print("\nCleaned DataFrame:\n", df)
```

Sample Output

Original DataFrame:

```
name marks
0  A  78.0
1  B   NaN
2  C  90.0
3  B   NaN
4  D  85.0
```

Missing Values:

```
name    0
marks   2
```

Cleaned DataFrame:

```
name  marks
0  A  78.000000
1  B  84.333333
2  C  90.000000
4  D  85.000000
```

Result

Thus, **missing values were identified and handled, and duplicate records were successfully removed using Pandas.**